SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 10

Code of Conduct
I Shri Shanmuga Priya K (192324035) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Assessment Tasks
Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

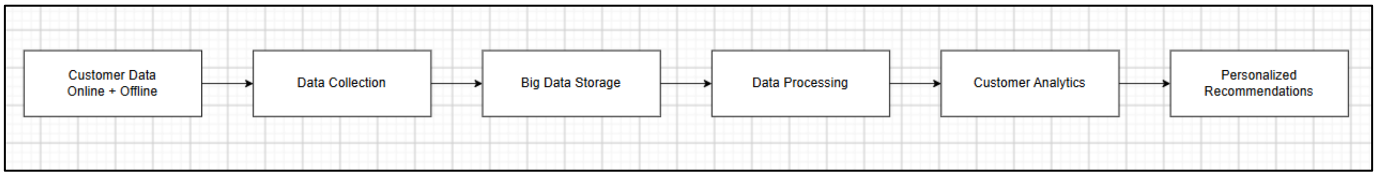
1. Unified Big Data Architecture for Multinational Retail Customer Analytics

Introduction

A multinational retail organization collects customer data from many sources such as websites, mobile applications, physical stores, loyalty programs, social media, and payment systems. When this data is stored separately, it creates **fragmented customer information**, making it difficult to understand customer behavior.

A unified big data architecture can integrate all these sources and provide **real-time analytics and personalized recommendations**.

Architecture



Components

1. **Data Sources**
    - Online purchases
    - Physical store transactions
    - Customer profiles
    - Loyalty cards
    - Website clicks
    - Mobile app activity
    - Social media interactions
2. **Data Ingestion**
    - Apache Kafka can collect real-time events.
    - APIs and ETL tools can collect batch data.
3. **Data Lake**
    - Stores structured, semi-structured and unstructured data.
    - Cloud storage such as Amazon S3 or Azure Data Lake can be used.
4. **Processing**
    - Apache Spark processes large datasets.
    - Apache Flink can process continuous real-time streams.

5. **Customer 360°**
   Data from different channels is combined into a single customer profile containing:
   - Purchase history
   - Preferences
   - Browsing behavior
   - Location
   - Loyalty information
6. **Recommendation Engine**
   Machine learning models analyze customer behavior and recommend relevant products.

Benefits

- Unified customer view
- Real-time customer analytics
- Personalized recommendations
- Improved customer experience
- Better marketing decisions
- Increased sales and customer retention

Conclusion

A cloud-based big data architecture using **Kafka, data lakes, Spark/Flink and machine learning** can eliminate data fragmentation and provide real-time personalized customer experiences.

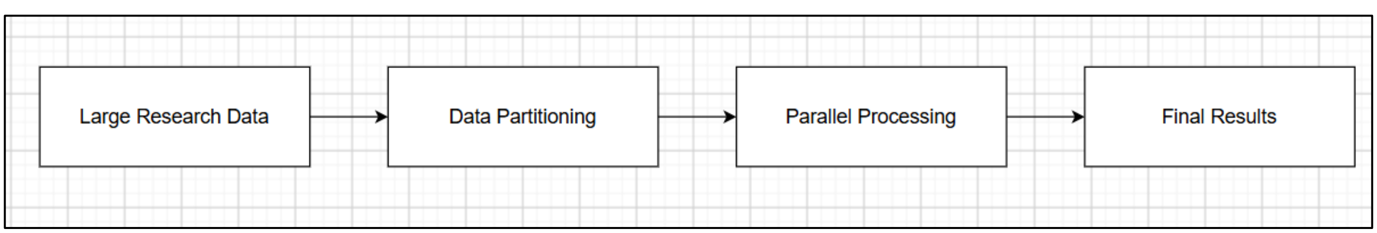
2. Distributed Computing for Large-Scale Scientific Research

Introduction

Scientific research generates enormous datasets from simulations, experiments, satellites, sensors, genomic analysis and scientific instruments. Processing such data using a centralized system is slow and difficult.

Distributed computing divides a large computational task among multiple computers.

Flowchart



Centralized vs Distributed Computing

Centralized System	Distributed System
Single powerful computer	Multiple computers
Limited processing capacity	High processing capacity
Processing is mostly sequential	Parallel processing
Single point of failure	Fault tolerance possible
Difficult to scale	Easily scalable
Large processing time	Reduced processing time

How Parallel Processing Improves Efficiency

Suppose a dataset contains **1 billion records**. A single computer must process all records sequentially. With 100 processing nodes:

1 Billion Records → 100 Partitions → 100 Nodes Process in Parallel → Results Combined

Ideally, the workload is divided among the nodes, significantly reducing processing time.

Technologies

- Hadoop
- MPI
- Kubernetes
- Cloud computing
- GPU clusters

Advantages

1. **Speed** – Multiple tasks execute simultaneously.
2. **Scalability** – Additional machines can be added.
3. **Fault tolerance** – Failed tasks can be restarted on other nodes.
4. **Large data processing** – Suitable for TB/PB-scale datasets.
5. **Resource utilization** – Uses multiple CPUs/GPUs efficiently.

Conclusion

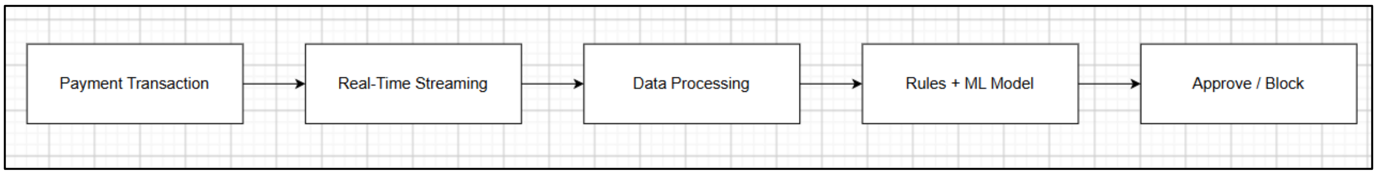
Distributed computing provides better **performance, scalability and fault tolerance** than centralized computing for large-scale scientific research.

3. Big Data Fraud Detection Framework for Digital Payments

Introduction

Digital payment platforms generate huge numbers of transactions every second. Fraud must be detected quickly before a transaction is completed. A combination of **real-time streaming, big data analytics, machine learning and rule-based detection** can provide effective fraud prevention.

Architecture



Important Features

The system analyzes:

- Transaction amount
- Transaction frequency
- Location
- Device information
- IP address
- Merchant information
- Previous transaction behavior
- Time of transaction

Rule-Based Detection

Example:

IF transaction_amount > threshold

AND location is unusual

AND transaction_frequency is high

THEN increase fraud score

Machine Learning

Machine learning models can identify complex fraud patterns.

Possible models:

- Random Forest
- XGBoost

- Logistic Regression
- Neural Networks
- Isolation Forest

Why Streaming Processing?

Traditional batch processing may identify fraud **after several minutes or hours**. Streaming processing analyzes transactions immediately.

Benefits

- Real-time fraud detection
- Reduced financial losses
- Faster response
- Improved accuracy
- Continuous model improvement
- Detection of new fraud patterns

Conclusion

A hybrid framework combining **stream processing + rules + predictive machine learning** provides faster and more accurate fraud detection than traditional batch-based systems.

4. Storing and Processing Multimedia Data in Big Data

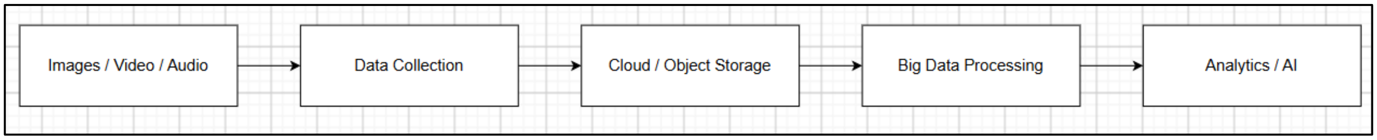
Introduction

Multimedia data includes **images, videos and audio files**. These datasets are large, complex and usually unstructured. For example, a high-quality video can occupy several GB of storage. Millions of videos therefore require petabytes of storage.

Challenges

1. Huge storage requirements
2. High bandwidth requirements
3. Different file formats
4. Unstructured nature, Complex processing
5. Metadata management
6. Fast retrieval requirements
7. Data security

Architecture



Recommended Technologies

Requirement	Technology
Object storage	Amazon S3 / Azure Blob
Distributed storage	HDFS
Processing	Apache Spark
Streaming	Kafka
Metadata	NoSQL databases
Video processing	FFmpeg
Data analysis	Spark
Content delivery	CDN

Storage Mechanisms

Object storage is preferred because it provides:

- High scalability, High durability
- Easy access, Low storage cost
- Support for huge files

Data Processing

Spark can process metadata and multimedia-related datasets in parallel. AI models can be used for:

- Image recognition, Speech recognition
- Video classification
- Object detection
- Sentiment analysis

Conclusion

A combination of **cloud object storage, distributed processing, NoSQL metadata storage and CDN technology** provides an efficient solution for multimedia big data.

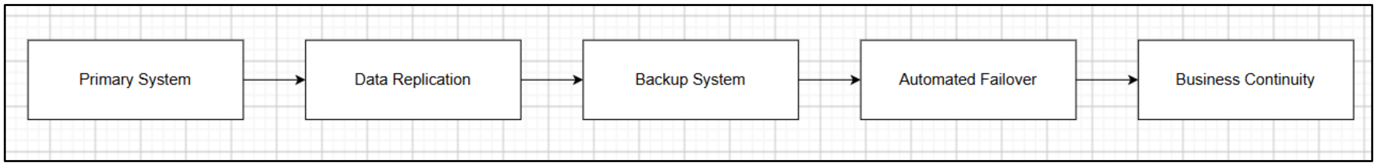
5. Disaster Recovery and Business Continuity for Cloud Big Data

Introduction

A cloud-based big data platform may face hardware failures, network failures, cyberattacks, software failures or natural disasters.

Therefore, organizations need **Disaster Recovery (DR)** and **Business Continuity Planning (BCP)**.

Architecture



Key Strategies

1. Redundancy

Multiple servers and resources are maintained. If one server fails:

Server A → Failed

↓

Server B → Continues Service

2. Replication

Data is copied to another location. Replication can occur:

- Within a region
- Across regions
- Across availability zones

3. Automated Failover

When the primary system fails, traffic automatically moves to the backup system.

4. Backup

Regular backups protect against Accidental deletion, Data corruption, Cyberattacks.

5. Recovery Objectives

- RPO – Recovery Point Objective: Maximum acceptable data loss.
- RTO – Recovery Time Objective: Maximum acceptable downtime.

Benefits

- High availability

- Reduced downtime
- Reduced data loss
- Improved reliability
- Business continuity

Conclusion

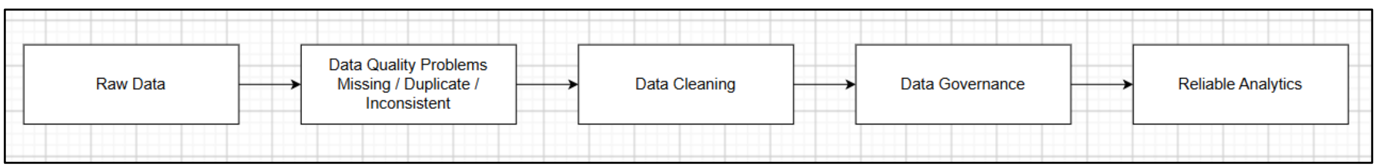
Using **redundancy, replication, backups and automated failover** ensures that a cloud-based big data platform can continue operating even during major failures.

6. Impact of Data Quality Issues and Data Governance

Introduction

Data quality is extremely important in big data analytics. Poor-quality data produces inaccurate analytical results and incorrect business decisions.

Common Data Quality Problems



1. Missing Values

Missing values can reduce analytical accuracy.

Solution:

- Mean/median imputation
- Prediction-based imputation
- Removing records when appropriate

2. Duplication

The same customer may appear multiple times.

Solution:

- Deduplication
- Unique customer IDs
- Master data management

3. Inconsistency

Example: India, IND, IN These may represent the same country.

Solution:

- Standard formats
- Data validation rules

Data Governance Strategies

1. Data quality rules
2. Data validation
3. Master Data Management
4. Data lineage
5. Metadata management
6. Data ownership
7. Access control
8. Regular data auditing

Data Quality Process

Data Collection → Data Validation → Data Cleaning → Deduplication → Standardization → Quality Monitoring → Reliable Analytics

Conclusion

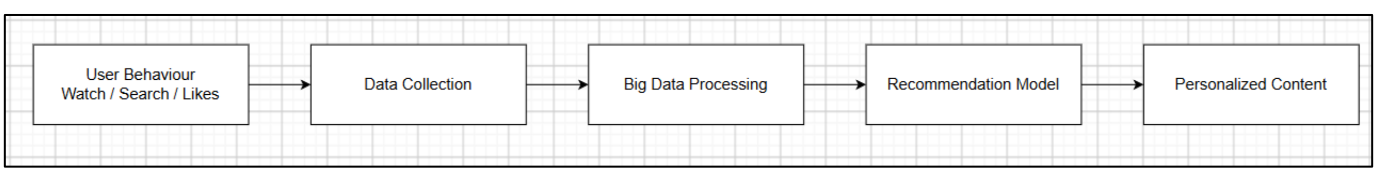
Strong data governance ensures that data is **accurate, consistent, complete, secure and trustworthy**, improving the reliability of big data analytics.

7. Scalable Recommendation System for Online Streaming

Introduction

Online streaming platforms such as video or music services have millions of users and content items. A recommendation system analyzes user behavior and recommends content that users are likely to enjoy.

Architecture



Recommendation Techniques

1. Collaborative Filtering

Recommends content based on similar users.

Example:

User A likes X, Y, Z. User B likes X, Y → Recommend Z to User B

2. Content-Based Filtering

Recommends content based on characteristics of previously watched content.

3. Hybrid Recommendation

Combines both approaches.

Big Data Technologies

- Kafka – real-time user events
- Spark – distributed processing
- NoSQL – user/content data
- Cloud storage – large-scale datasets
- ML models – recommendations

How It Improves Engagement

The system can analyze:

- What users watch, When they watch
- How long they watch
- What they skip
- Search behaviour, Likes/dislikes

Recommendations become increasingly personalized.

Benefits

- Higher user engagement
- Longer viewing time
- Better content discovery
- Improved customer retention
- Scalable to millions of users

Conclusion

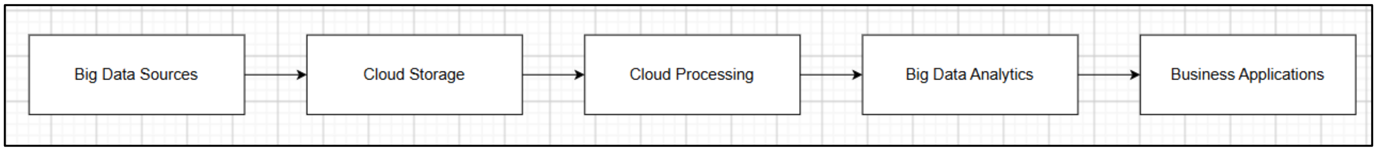
Big data technologies enable streaming platforms to process enormous amounts of user behavior data and generate **real-time, personalized recommendations**.

8. Role of Cloud Computing in Big Data

Introduction

Cloud computing provides on-demand computing, storage and networking resources. It has become an important foundation for modern big data systems.

Architecture



Advantages

1. Scalability

Resources can be increased when data volume increases.

Low Demand → Few Resources

High Demand → More Resources

2. Cost Management

Organizations can use a pay-as-you-go model rather than purchasing large amounts of hardware.

3. Flexibility

Cloud platforms provide:

- Storage
- Databases
- Analytics
- AI/ML
- Networking

4. Faster Deployment

Infrastructure can be created quickly.

Limitations

Security: Data breaches, Misconfigured storage, Unauthorized access

Cost: Uncontrolled resource usage can increase costs.

Vendor Lock-in: Migrating from one cloud provider to another can be difficult.

Network Dependency: Cloud applications depend heavily on network connectivity.

Compliance: Data location and privacy regulations must be considered.

Evaluation

Factor	**Cloud Computing**
Scalability	Excellent
Flexibility	Very high
Initial cost	Low
Long-term cost	Must be monitored
Security	Strong but configuration-dependent
Deployment	Fast
Maintenance	Lower than on-premises

Conclusion

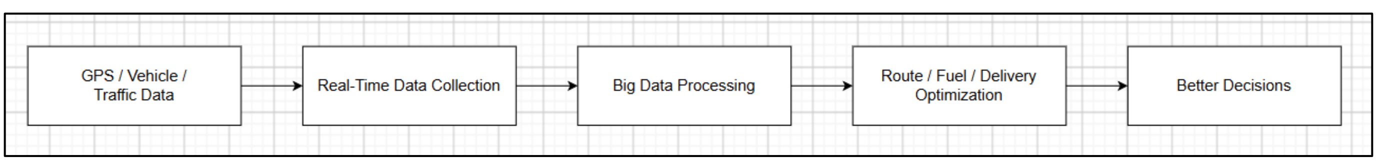
Cloud computing provides **scalability, flexibility and rapid deployment**, making it ideal for big data. However, organizations must carefully manage **security, cost, compliance and vendor dependency**.

9. Real-Time Analytics Framework for Transportation and Logistics

Introduction

Transportation companies generate real-time data from GPS devices, vehicles, traffic systems, weather services and delivery applications. Big data analytics can use this information to optimize routes, reduce fuel consumption and improve operational decisions.

Architecture



Applications

1. Route Optimization

Real-time traffic information can be analyzed to identify the fastest route.

Route A → Heavy Traffic → Avoid

Route B → Moderate Traffic → Select

2. Fuel Efficiency

Analytics can identify:

- Excessive idling
- High-speed driving
- Inefficient routes
- Vehicle maintenance issues

3. Predictive Maintenance

Vehicle sensor data can predict possible failures before they occur.

4. Delivery Optimization

The system can consider:

- Distance
- Traffic
- Delivery priority
- Vehicle availability
- Weather

Technologies

- IoT sensors
- Kafka
- Spark/Flink
- GPS
- Cloud computing
- Machine learning
- NoSQL databases

Benefits

- Reduced travel time
- Reduced fuel consumption
- Better fleet utilization
- Faster deliveries
- Lower operational costs
- Improved customer satisfaction

Conclusion

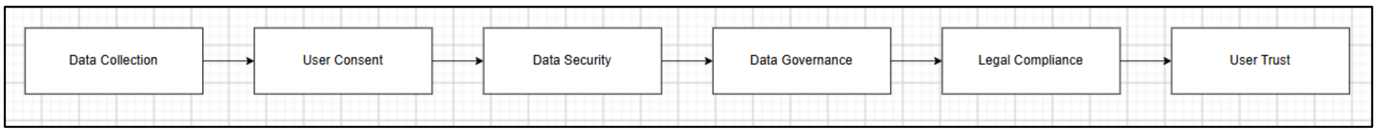
Real-time big data analytics allows transportation organizations to make **faster, data-driven decisions** while improving route efficiency and reducing operating costs.

10. Ethical and Legal Challenges of Big Data

Introduction

Big data allows organizations to collect and analyze enormous amounts of personal information. However, improper collection or use of data can violate privacy and reduce user trust.

Major Challenges



1. Privacy

Organizations may collect:

- Location
- Purchase history
- Browsing behavior
- Personal information

Users should know how their data is being used.

2. Lack of Consent

Data should not be collected or processed beyond what users have agreed to, subject to applicable laws and legal bases.

3. Security

Large databases are attractive targets for attackers. Security measures include:

- Encryption
- Authentication
- Access control
- Monitoring
- Data masking

4. Algorithmic Bias

Machine learning models can produce unfair results if training data contains bias.

Example:

Biased Data

↓

Biased Model

↓

Unfair Decision

5. Data Ownership

Organizations must establish who can access, use and share collected data.

6. Legal Compliance

Organizations must comply with applicable privacy and data-protection regulations, such as:

- GDPR
- India's Digital Personal Data Protection framework
- Sector-specific regulations where applicable

Recommended Organizational Policies

1. Privacy-by-design
2. Clear consent and privacy notices
3. Data minimization, Purpose limitation
4. Strong encryption, Role-based access control
5. Regular security audits
6. Data retention and deletion policies
7. Algorithmic fairness testing, Incident response procedures
8. Employee data-protection training
9. Regular compliance assessments

Responsible Big Data Framework

Data Collection, User Consent & Legal Basis, Secure Storage, Data Quality & Governance, Fair & Transparent Analytics, Controlled Data Sharing, Monitoring & Auditing, User Trust & Compliance

Conclusion

Big data should not be used only for business advantage. Organizations must balance **innovation with privacy, security, fairness and legal compliance**. Responsible data governance helps maintain both regulatory compliance and user trust.